Forward-Deployed Engineer | Case Study

Take-home build + short write-up

Thank you for your interest in the Forward-Deployed Engineer role at Habitat Health. This case is designed to mirror the actual work: taking a messy, real-world operational process and building something that makes it measurably better. We care far more about how you think, scope, and build than about polish. This case is scoped for about six hours. We would rather see a narrow thing that works than a broad thing that doesn't, so scope down before you push past that.

Business Context

Habitat Health partners with leading integrated care organizations to deliver the comprehensive suite of care delivery, care management, technology, and enrollment services to launch and operate high performing Programs of All-Inclusive Care for the Elderly (PACE) programs for eligible older adults. Habitat Health began serving participants in California in 2025, with a first center in Sacramento (2025), a second in South LA (2026), and several more planned to open in the next 18+ months.

PACE is easiest to think of as an alternative to being placed in a skilled nursing facility for those who are dually eligible for Medicare and Medi-Cal, with complex and often interwoven medical, behavioral, and social needs. The program allows people to remain in their homes with access to a comprehensive set of services at a center in their community: transportation, behavioral and physical health care, culturally-curated meals, socialization, and family support. PACE organizations both deliver a majority of a participant's care directly and are financially responsible for all of it, aligning patient outcomes and financial incentives.

How PACE Enrollment Works Today

Enrollment in California is a multi-step, time-sensitive process that blends centralized phone-based engagement with required in-person interactions. Each step depends on timely outreach and clean handoffs. The full process averages about 40 days.

- **Referral & Outreach:** prospects are referred by partners (e.g., Kaiser Permanente), family, or community outreach. Reaching them usually takes multiple call attempts.
- **Welcome Call:** an Enrollment Specialist (ES) or Outreach Specialist (OS) explains the PACE model, runs screening questions, and confirms understanding of the "lock-in provision."
- **Next Steps:** based on interest, prospects may be offered a nurse home visit (clinical assessment), an ES/OS home visit, or a center tour; the latter two aim to convert to a nurse visit.
- **Clinical Assessment (in-person):** a nurse conducts a home visit to determine eligibility and submits the Level of Care application to the California Department of Health Care Services (DHCS).
- **Enrollment Conference (in-person):** if approved, the ES/OS reviews the Participant Handbook and signs enrollment documents. Enrollment begins the first of the following month.

Prospects may also have friendly visits to build rapport, follow-up calls for reminders or document collection, additional center tours, and additional clinical assessments to support the Level of Care application.

We have many current technology systems that enable the team: Salesforce as the CRM, Docusign for prospective participant electronic signatures, Propio for live translation, Epic for clinical assessments, Microsoft 365 for internal communications and shared details across cross-functional staff, and Talkdesk as the uniform dial-out platform.

The Operational Reality

Today, each prospect is managed end to end by a single ES or OS who handles all phone outreach, follow-up, scheduling, and in-person engagements. Timely phone work constantly competes with field responsibilities, and availability swings with PTO, documentation load, travel, and surges or dips in referrals. As a result, referrals can wait for first contact, backlogs build quickly, and frequent task-switching hurts consistency. The net effect is an uneven prospect experience, lower conversion, and less predictable enrollment across the funnel.

The friction shows up in a few places: referrals wait too long for a first call and some go cold before anyone reaches them; referrals arrive from many sources with thin, inconsistent information; how reps run calls varies, so quality depends on individual memory rather than a shared standard; and work passes between roles with heavy manual data entry along the way. The same prospect shows up across the referral file, Salesforce, the call log, and the state forms, and no single view reconciles them, so it is easy for one system to say something another contradicts.

Your Task

Pick one problem worth solving and ship a small working prototype with a short README. You will not have access to Habitat's live Salesforce, Epic, Talkdesk, Outlook, or data warehouse, but we've included a synthetic dataset so you can start building immediately. We care more about your judgment on what to build than whether you picked the "right" area; tell us what you'd leave out, and why.

Whatever you build, your README and walk-through should cover:

1. **Problem & scope:** What problem did you choose to solve and why?
2. **Solution design:** How does your prototype work, and where did you use deterministic logic vs. AI? What integration points and assumptions did you make?
3. **Success criteria:** How would you measure impact and how would this hold up as it scales to additional centers?

Deliverables

Send a zip file or git repo containing the two items below. We should be able to open it and get your prototype running from the README alone.

1. Your code

- Runs with one documented command against synthetic or mocked data. A container is welcome but not required - a local setup is fine.
- Actually works on a happy path, even if narrow. A screenshot is not a substitute for something that runs.
- Use whatever language or framework you are most productive in. For context, our stack is primarily Python, with Databricks as our data platform. You do not

need to match this, but if running your prototype needs more than a standard Python setup, include a Dockerfile or note the dependencies so we can run it from the README alone.

2. README.md

- **How to run it** — the one command, and the input or request that produces output.
    - If your prototype calls an external AI service, it must also run in a demo mode that works without an API key, using pre-saved or mocked responses, so we can run it from the README alone. A live mode that reads a key from an environment variable is a welcome option, but the default run command must not require one. Note in the README which mode is which.
- **Your write-up –** address questions in "Your Task" above.

Using AI coding assistants is fine and expected. If it helped you build, be ready to explain and defend your choices — no need to submit your prompts.

The Walk-Through

You will present live after submitting: about 20 minutes to walk us through your thinking and demo the prototype, then about 10 minutes of discussion. Slides are optional. This discussion will take place virtually, via a Microsoft Teams link that will be sent to you in a meeting invite from Greenhouse, our scheduling portal.

- **Use a language or framework you are comfortable with**, including AI coding assistants.
- **Send your submission** (repo or zip) at least 24 hours before your scheduled walk-through.

If you have questions or hit a blocker, please reach out to Gelsey Azhderian, Talent Acquisition, who will coordinate with the internal team.